Evidence Availability and Retrieval in Scientific Claim Verification: A Matched SciFact Development-Set Study

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Abstract

To verify scientific claims, the appropriate evidence must be collected and carefully analysed. This study compares three evidence-input conditions on a matched subset of the SciFact development split: claim only (A), claim plus annotated gold evidence (B), and claim plus the top five abstracts retrieved using BM25 (C). For each condition, a separate TF-IDF and logistic regression classifier is fitted using the same training subset and condition-specific inputs. The input of each classifier is condition-specific. The preserved assessment contains 286 aligned development assertions, with annotated gold evidence for 182 assertions. The macro-F1 scores for A, B, and C over all 286 claims are 0.3757, 0.4988, and 0.3650, respectively. The apparent advantage of B on the full dataset is influenced by NOT ENOUGH INFO claims with null evidence, creating a shortcut based on evidence existence. In the 182-claim gold-evidence segment, B achieves a macro-F1 of 0.3492 compared to 0.3140 for A; the paired difference is +0.0352 (95% bootstrap CI: -0.0275 to 0.0974), and does not meet the enhancement criterion. BM25 achieves a claim-level Hit@5 of 90.11%, while TF-IDF cosine retrieval achieves 82.97% on the same measure. The BM25-based classifier is not able to beat the claim-only model on the evidence segment. We performed a taxonomy-directed evaluation on a sample of 50 cases. The 50 sampled error cases were reviewed using the predefined taxonomy, and their classifications were verified against the corresponding experimental records and the exact historical local corpus snapshot. The results show that retrieval success and correct classification of scientific claims are separate outcomes, and that the evaluated lexical verifier performs poorly, in particular on CONTRADICT claims.

Keywords

SciFact, scientific claim verification, evidence retrieval, BM25, TF-IDF, logistic regression, claim-level Hit@5, macro-F1

1 Introduction

Scientific claims are often required to be backed up by specialist academic journals. A verifier must be able to find relevant research abstracts and determine whether their conclusions support, contradict, or are inconclusive of a claim. SciFact formalises this process with scientifically authored claims, abstracts that include evidence, and rationale annotations [1]. The framework allows for analysis of two related but different processes: collection of evidence and making a judgement once evidence has been identified.

These operations are not to be regarded as synonymous. For example, a retrieval system may return a relevant document, but the following classifier may not be able to recognise the relationship between the document and the assertion. Or a classifier might appear to perform well by taking advantage of patterns in the input that are correlated with labels rather than doing any scientific work. Therefore, the evidence existence, the retrieval quality and the classification performance must be evaluated separately.

In this work, we experiment with three evidence-input settings on a matched SciFact dev set: (A) claims-only input, (B) claims + annotated gold evidence input, and (C) claims + top-5 BM25-retrieved abstracts input. We use TF-IDF features and logistic regression as it is a light-weight baseline and efficient on the CPU. The study evaluates existing evidence-input strategies rather than proposing a new verification paradigm. The goal is to perform an auditable assessment of condition-specific pipelines. Existing work has studied retrieval-augmented verification of scientific claims, such as CliVER [2].

The study makes four contributions. It first presents a matched comparison of the three input conditions for each claim. Second, it shows retrieval efficacy regardless of the judgement category, with a claim-level Hit@5 metric. Third, it looks into a structural bypass in the gold-evidence scenario due to a lack of evidence for NOT ENOUGH INFO assertions. Finally, it discusses the limitations of the archived run and reports the completed taxonomy-based review of sampled errors. These contributions are intended to support careful interpretation and reproducibility rather than to introduce a new benchmark.

2 Related Work

Wadden et al. (2018) proposed SciFact, a scientific claim verification benchmark that requires systems to retrieve evidence and justify the arguments that support or refute the claims. The standard promotes the separation of document retrieval and verdict prediction . Liu et al. proposed CliVER, a full retrieval-augmented framework for verifying scientific claims, and performed label prediction evaluation on SciFact and CoVERt. In addition, they evaluated Open and Oracle settings for an independent clinician-claim task [2]. This study, therefore, does not claim originality in comparing gold and retrieved evidence. The emphasis is on the aligned evaluation documentation, the identification of a vacuous-evidence shortcut, and the careful appraisal of a streamlined lexical validator.

There are a number of neighbouring assessment frameworks that help explain why you would want to report evidence retrieval and then accuracy separately. FACTSCORE decomposes large amounts of generated text into atomic facts and quantifies the fraction supported by a source of knowledge [3]. ALCE evaluates citation-based generation by criteria on accuracy and quality of citations [4]. RAGAs separates the evaluation of the retrieved context from the evaluation of the generated response in retrieval-augmented generation [5]. These frameworks encourage detailed support analysis, but their tasks and metrics are not numerically substitutable for three-way SciFact classification.

RAV focuses on the significance of retrieval accuracy for fact verification on FEVER [6]. Chen et al. investigate the gathering of

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genuine evidence for intricate political assertions through temporary online evidence [7]. Two preprints in 2026 explore retrieval and taxonomy-oriented error diagnosis in biomedical or medical factuality contexts [10, 11]; these are considered here as nascent relevant research, and not established benchmarks. BM25 is a lexical retrieval method in the probabilistic relevance framework [8]. TF-IDF and logistic regression are implemented using the Scikit-learn framework [9].

3 Research Questions and Decision Rule

The study addresses the following research questions:

- RQ1: How do claim-only (A), claim with annotated gold evidence (B), and claim with retrieved evidence (C) compare in three-way verification performance on the same eligible SciFact development claims?
- RQ2: What is the frequency that the top-five retrieval has at least one annotated gold evidence document? How is the retrieval hit/miss status descriptively associated with subsequent accuracy?
- RQ3: What are the pre-defined error classifications for each condition? A stratified sample of 50 error cases was examined using a predefined taxonomy and static codebook, with each classification verified against the corresponding condition-specific inputs and the exact historical local corpus snapshot.

The main metric for RQ1 is Macro-F1 which gives equal importance to the classes, regardless of their uneven frequencies. The pre-defined practical-effect reference shows a total macro-F1 gain of +0.03. An improvement is considered validated when the expected increase is at least +0.03 and the 95% paired-bootstrap confidence interval does not contain zero. The expectation that B would do better than A and C would be close to B are treated as problems to study, not as findings.

4 Dataset and Data Construction

4.1 Dataset partitions

The audit of archived data consists of 809 labelled training claims, 300 development claims and 5,183 corpus documents. The benchmark release's 300 claims without public labels were in the sanctioned test division and were not evaluated for metrics. The authoritative repository details the files for training, development, and testing, and sends test evaluation to the benchmark leaderboard [1]. Thus, all results in this paper are on the development set and we do not make any claims about the test set or leaderboard performance.

4.2 Matched evaluation set and duplicate filtering

Fourteen development claims with near-duplicate training counterparts were held out using a token-set Jaccard similarity threshold of at least 0.90. This left 286 claims shared across the three evaluation conditions. Their class distribution is 60 CONTRADICT, 104 NOT ENOUGH INFO, and 122 SUPPORT. Of these, 182 claims have annotated gold evidence: 60 CONTRADICT and 122 SUPPORT. The evidence-bearing subset therefore contains two classes and no NOT ENOUGH INFO examples. A sensitivity analysis using a stricter near-duplicate similarity threshold of 0.70 leaves 209 matched claims, of which 127 have annotated gold evidence.

4.3 Label mapping and evidence construction

The archived label mapping considers an absent evidence annotation as INSUFFICIENT INFORMATION, a single SUPPORT or CONTRADICT annotation as the corresponding label and a missing evidence key as an unlabelled entry. Entries whose mappings are not clear or not specified are omitted. The three criteria are not met until the claim and document identifiers are aligned. Some claims have no annotated gold evidence . These claims are not counted as retrieval failures . These claims are excluded from the basis of the retrieval scoring . This mapping has been captured in the project audit results and needs to be cross-checked with the exact raw dataset schema for any new acquisition.

The input for condition B is the initially annotated rationale set for each evidence document in the order of the files, as per the documented protocol amendment. An independent audit has found that 75 of 182 evidence-eligible matched claims have multiple rationale sets in at least one document. The choice of the initial rationale set is an important operational decision. The findings should not be generalised to union-of-rationales or sentence-level evidence synthesis without a new ablation.

5 Experimental Protocol

5.1 Evidence-input conditions

Condition A only receives the scientific statement. Condition B gets the claim along with the initial rationale set annotated for each gold evidence document, in file order. B lacks gold evidence for the claims due to insufficient information, so B is merely an oracle-style evidence condition for the 182 claims subset that has annotated gold evidence. Condition C takes the claim, title and abstract text from the five most relevant documents retrieved by BM25. Gold document identifiers and labels are only used for evaluation after retrieval; they are not used to rank candidate documents.

5.2 Classification model and training controls

Each condition is developed as an independent TF-IDF and logistic regression model. All models use the same identifiers for training claims, the same algorithm family, the same random seed, and the same core hyperparameters. However, the inputs to their training correspond to the particular condition. The vectorizer uses word unigrams and bigrams, with minimum document frequency of 1, L2 normalisation and a maximum of 100,000 features. Logistic regression is trained with C = 1, lbfgs optimiser, 2000 iterations max, no class weighting, and argmax for predictions. The lexicon is customised only to the training resources. The order of classifications is CONTRADICT, INSUFFICIENT INFORMATION and SUPPORT.

This design differs from comprehensive, condition-specific workflows. It does not keep one calibrated classifier and only changes the evidence given in the evaluation phase. So the noted discrepancies could be changes in the training representation as well as changes in the evidence available at inference. The study does not interpret

these differences to mean that evidence availability alone has a causal effect.

5.3 Retrieval methods

BM25 is run with rank-bm25 version 0.2.2, using k1=1.5, b=0.75 and epsilon=0.25. Queries are lower-cased and tokenised using the regular expression for words (
w+). No removal of stop-words or stemming has been performed. The five top ranked papers are merged based on their titles and abstracts. Retrieval scores are sorted in increasing numerical order of document identifiers. A TF-IDF cosine retriever is an auxiliary retrieval comparator. Both retrievers are tested on the same corpus and query set.

5.4 Evaluation metrics and uncertainty

The classification is evaluated in terms of accuracy, macro-F1, precision and recall per class, F1 scores and confusion matrices. We generated 2,000 claim-level bootstrap resamples using seed 42. 95% percentile confidence intervals are used to assess the paired differences B-A, C-A and B-C. Pairs of conditions within each resample are matched by claim identifier.

Retrieval is evaluated on claim-level Hit@5 on the 182 claims with annotated gold evidence. A claim is successful if at least one of its designated gold evidence documents is among the five retrieved documents. Thus Hit@5 is the proportion of qualifying claims with at least one gold-document hit. It is not about document level recall, which means counting the relevant documents retrieved out of the total number of relevant documents.

6 Audit and Protocol Validation

6.1 Rationale for the development-set evaluation

The official development subset is used in the course project, as the benchmark release does not provide the designated test subset publicly. The exam is presented as a clear evaluation of the development set. The main set of matches uses a 0.90 near-duplicate threshold and keeps 286 claims; the 0.70 sensitivity analysis changes both the sample size and the examples included. The two sets of results are therefore not direct reproductions of the same claims.

The full set of 286 claims has all three labels, but the gold-evidence subset of 182 only has SUPPORT and CONTRADICT. So its macro-F1 is an average of two classes, unlike the three-class macro-F1 across all matched claims. In the document these segments are kept distinct when analysing outcomes.

6.2 Internal consistency checks

The independently conducted audit with timestamps shows 12 PASS evaluations, four WARN evaluations, two INFO evaluations and no FAIL evaluations. The verifications are: checking the number of train and dev labels match the SciFact publication, checking the corpus size, removing wrong evidence-document citations, recalculating metrics from stored predictions, automatic re-training under condition A, vocabulary adaptation only on train data, independent BM25 validation, re-calibrating retrieval Hit@5, and verifying the configuration hash matches the results hash. These

Table 1: Saved development-set classification results. The 286-claim slice is three-class; the 182-claim evidence slice is two-class.

Evaluation slice	Condition	Accuracy	Macro-F1
All matched (n = 286)	A: Claim only	0.4615	0.3757
	B: Gold evidence	0.6014	0.4988
	C: BM25 top-5	0.4371	0.3650
Gold-evidence (n = 182)	A: Claim only	0.3901	0.3140
	B: Gold evidence	0.4835	0.3492
	C: BM25 top-5	0.3736	0.3119

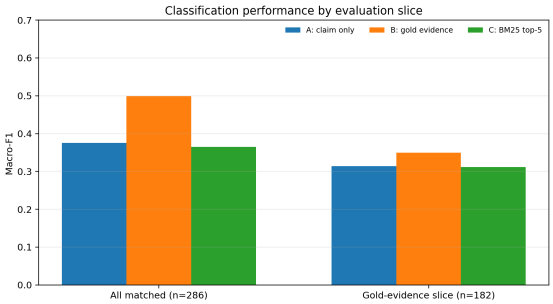


Figure 1: Macro-F1 by evidence-input condition. The two evaluation slices contain different class sets and should not be compared as identical estimands.

checks improve the internal consistency of the archived documents, but they do not check that every modelling choice reproduces the official SciFact framework.

Condition independent refit The maximum absolute difference between the probabilities and the stored probabilities was about 5×10^{-9}, which is consistent with the expected effect of storing probabilities to eight decimal places. Independent vocabulary check detected 9,018 terms when fitting on training data only and 10,957 terms when training and development data were combined. This examination would identify the contribution of the lexicon of development data.

7 Results

7.1 RQ1: classification performance

Table 1 Summarises results of retained assessments. In the matched set of 286 claims, the highest macro-F1 score is for condition B (0.4988), followed by A (0.3757) and C (0.3650). In the 182 claim gold evidence section, B gets a macro-F1 score of 0.3492 while A gets 0.3140 and C gets 0.3119. The accuracy maintains a similar general ordering for the B condition, but the overall claim outcome is influenced by the empty-evidence heuristic described later.

The B-A macro-F1 difference for the first 286 claims is +0.1231 (95% C.I. 0.0721 to 0.1783). However, this quantitatively favourable disparity can not be interpreted as a gold-standard advantage since the empty evidence domain is the NOT ENOUGH INFO category in the B condition. The evidence-bearing segment is a more refined

Table 2: Per-class F1 on all 286 matched development claims.

Class	A: Claim only	B: Gold evidence	C: BM25 top-5
CONTRADICT	0.0476	0.0444	0.0659
NOT ENOUGH INFO	0.5865	0.8442	0.5616
SUPPORT	0.4929	0.6078	0.4676

comparison for claims supported by gold evidence, but it only has two categories.

The B-A difference in that 182-claim segment is +0.0352 (95% CI: -0.0275 to 0.0974). The pre-established criterion is not met because, although the point estimate exceeds the +0.03 practical-effect benchmark, the interval includes zero. The C-A difference was 0.0021 (95% CI 0.0629 to 0.0591) and was not conclusive.

7.2 Per-class performance

The per-class F1 scores for all 286 corresponding claims are presented in Table 2. The largest deficiency is in the CONTRADICT class: F1 scores are 0.0476 for A, 0.0444 for B, and 0.0659 for C. Conversely, the SUPPORT F1 is 0.4929, 0.6078, and 0.4676 respectively. Condition B achieves a strong NOT ENOUGH INFO F1 score of 0.8442, consistent with its ability to exploit the empty-evidence pattern. This score should not be construed as evidence-based rationale.

7.3 Empty-evidence shortcut diagnostic

A deterministic rule that predicts NOT ENOUGH INFO if there is no evidence for condition-B and SUPPORT otherwise has an accuracy of 0.7902 and a macro-F1 of 0.6009 on the whole set of 286 claims. Both measures outperform condition B on the full dataset (accuracy 0.6014; macro-F1 0.4988). This simple principle uses the existence of evidence as a proxy for the label and hence the all-claim B comparison is inherently confounded.

The same principle predicts SUPPORT for every claim in the 182-claim gold-evidence subset, with an accuracy of 0.6703 and a macro-F1 score of 0.4013, again beating condition B (accuracy 0.4835; macro-F1 0.3492). The difference in macro-F1 for the B-minus rule on this segment was 0.0521 (95% CI 0.0927 to 0.0057). The results suggest caution is needed in interpreting the performance in condition B as evidence of the classifier having learned how to interpret scientific evidence.

7.4 RQ2: retrieval and downstream correctness

BM25 finds at least one annotated gold evidence document for 164 of 182 eligible claims, yielding a claim-level Hit@5 of 90.11%. TF-IDF cosine retrieval identifies at least one annotated gold evidence document for 151 of the 182 eligible claims, corresponding to a claim-level Hit@5 of 82.97%. These metrics indicate whether a claim is found at least once in the gold documents among the top 5 retrieved documents, but do not measure the fraction of all retrieved gold documents.

BM25 performs better on Hit@5 but condition C is not better than claim-only condition. In the evidence section with 182 claims, the accuracy of C is 0.3736 and the accuracy of A is 0.3901. Table 3 presents the analysis of hits and misses. Of the 164 hit claims made

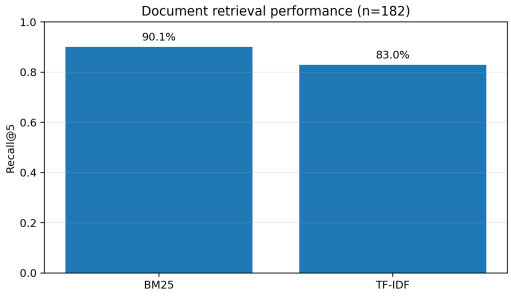


Figure 2: Claim-level Hit@5 over the 182 claims with annotated gold evidence. A hit requires at least one annotated gold document among the top five retrieved documents.

Table 3: Descriptive accuracy by BM25 hit/miss status on the 182-claim evidence slice.

BM25 status	n	Class composition	A accuracy	B accuracy	C accuracy
Hit	164	115 SUPPORT; 49 CONTRADICT	0.4024	0.5000	0.3963
Miss	18	7 SUPPORT; 11 CONTRADICT	0.2778	0.3333	0.1667

Table 4: Paired macro-F1 differences on the 127-claim sensitivity evidence slice.

Sensitivity analysis (Jaccard 0.70)	Estimate	95% bootstrap CI
B-A macro-F1 difference	+0.0564	-0.0401 to 0.1535
C-A macro-F1 difference	-0.0037	-0.0901 to 0.0852

by BM25, 115 are labelled as SUPPORT and 49 as CONTRADICT, while of the 18 miss claims, 7 are labelled as SUPPORT and 11 as CONTRADICT. The accuracies are group-dependent but the small miss group and heterogeneous class composition make the comparison more descriptive than causal.

The recorded Fisher exact test for condition C hit versus miss has $p = 0.0723$. Given the 18-claim miss group and the descriptive nature of this subgroup comparison, the result is not treated as a definitive hypothesis test or as evidence of a causal relationship.

7.5 Sensitivity to the near-duplicate threshold

For a token-set Jaccard threshold of 0.70, the sensitivity analysis includes 209 corresponding claims and 127 claims with evidence. In the subset of patients with evidence, B-A is +0.0564 (95%CI: -0.0401 to 0.1535) C-A is -0.0037 (95%CI: -0.0901 to 0.0852). Both intervals contain zero. These projections are consistent with the main findings from the evidence-slice in terms of direction, but remain uncertain. This is a sensitivity analysis, as the threshold that affects the distribution of the incorporated claims and classes has been modified, so that the same sample cannot be reproduced.

8 Discussion

The results confirm three major observations. Condition B appears to be much more robust in the full three-class matched set, but this comparison is complicated by the lack of evidence about NOT ENOUGH INFO assertions. The B classifier performs worse than a deterministic evidence-presence criterion over the whole dataset,

so it is incorrect to consider the overall improvement as progress in evidence interpretation. Second, for statements with annotated gold evidence, the point estimate of B-A is positive, but the confidence interval associated with it contains zero. The established standard for a reliable practical improvement is therefore not met. Third, BM25 retrieves at least one labelled gold document for more eligible claims than TF-IDF cosine retrieval, but this benefit does not translate into improved classification accuracy for condition C compared to claim-only condition A.

The surprisingly low CONTRADICT F1 across all three scenarios is consistent with a limitation of bag-of-words models. One possible explanation is that the lexical features are not good at capturing negation, directionality or relational context. However this mechanism is not disentangled by the present experiment and therefore should be taken as a plausible interpretation rather than a scientifically verified cause.

The other possible problem is the amount of text given for condition C. The project audit shows the average of the retrieved input combined is approximately 8,000 characters and presumes the limit of 100,000 features is restrictive. The length of input and the degree of feature saturation could influence the performance, but the recorded experiments did not distinguish the separate effects of the two. A regulated ablation would be needed before asserting a causative relation for either element.

Overall, the results lead to a cautious negative or ambiguous conclusion rather than a performance assertion. Retrieval can help identify relevant documents (measured by claim-level Hit@5), but the lexical classifier may still fail to use these documents to infer the correct conclusion. Therefore, the retrieval quality and the label accuracy should be evaluated and reported separately.

9 RQ3: Taxonomy-Based Error Analysis

After finalising the seven-category codebook, a stratified sample of 50 erroneous claim-condition predictions was extracted, with one modification to the codebook documented for claims without gold evidence. The sample consists of 17 errors from A, 17 from B and 16 from C, selected with seed 42. Each case was reviewed against the claim text, the relevant gold or retrieved abstract text, the corresponding condition-specific inputs, and the exact historical local corpus snapshot. AI assistance was used during the initial coding process, and the resulting classifications were subsequently checked and verified by the student.

The reviewed sample comprises 31 LABEL DECISION ERROR assignments, 14 AMBIGUOUS ANNOTATION assignments, three OTHER UNCERTAIN assignments, one NO EVIDENCE SEEN assignment, and one RETRIEVAL MISS assignment. The two metadata-deterministic assignments are B/303 (a no-evidence case) and C/756 (a retrieval-failure case); the other 48 are text-review evaluations. These counts describe the reviewed 50-case sample.

The student reviewed and verified all 50 classifications against the exact historical local corpus snapshot and the corresponding condition-specific inputs. The reported counts describe only the selected sample; they are not prevalence estimates for the entire SciFact dataset. A separate second annotator was not employed, so inter-annotator agreement and Cohen's kappa are not reported. Raw abstracts are not distributed; the project archives source URLs, identifiers, review notes, and the versioned codebook and coding records.

10 Limitations and Threats to Validity

The research relies on a single development split, and a single frozen run, rather than an unmodified labelled test set or multi-seed evaluation. The matched set is not as comprehensive as the full development split and contains 286 claims after near-duplicates are removed. The three conditions are separate fits of a common model family. The design does not separate out a causal intervention on one fixed model. Furthermore, the evidence shortcut is missing in condition B, which makes the comprehensive set comparison more difficult. And the evidence informing part consists of only two categories.

The lexical verifier has especially low CONTRADICT F1 across all conditions. The BM25 miss group contains only 18 claims, limiting subgroup analysis. Effect of the feature cap . BM25 epsilon for common terms . The Condition-B evidence framework identifies initial rationale sets for each document, and 75 of the eligible claims for evidence have multiple rationale sets in at least one document. The outcome might vary given the union-of-rationales vs. sentence-level aggregation method.

The RQ3 analysis is based on a stratified sample of 50 claim-condition errors rather than the complete evaluation dataset. The analysis is limited by the sample size and the absence of an independent second annotator. Consequently, inter-annotator agreement and Cohen's kappa are not reported, and the observed classification frequencies should not be generalised to the entire SciFact dataset.

The preserved project documentation does not allow for a reproduction of a full clean-checkout. No confirmation of execution of Colab/GPU. The original environment snapshot included Python 3.11.2, a CPU-only version of Torch, and CUDA was not available. The autonomous evaluation container was running Python 3.13.5 and did not have rank-bm25 installed, so the two testing modules that required retrieval could not be retrieved. Section 12 elaborates the constraints regarding the environment and testing.

The official license file for the SciFact repository finally specifies CC BY 4.0 for claim/evidence annotations, ODC-By 1.0 for corpus abstracts, and Apache 2.0 for code [12]. However, the source obtained was a dynamic release/latest archive rather than a fixed commit, and a Hugging Face mirror has a different CC BY-NC 2.0 license designation and a different split arrangement. It is still not clear how the downloaded archive corresponds to an upstream revision. We still exclude raw data until we confirm the license at the snapshot level.

11 Conclusion

We evaluate claim-only, annotated gold-evidence, and BM25-retrieved-evidence inputs on the SciFact development set using separately fitted TF-IDF and logistic-regression classifiers. The gold-evidence criterion obtains the highest macro-F1 score on the entire 286-claim dataset, but is confounded by empty-evidence NOT ENOUGH INFO cases, and outperformed by a simple evidence-presence rule. The observed B-A macro-F1 difference for the 182-claim gold-evidence segment is +0.0352, but its 95

BM25 achieves a claim-level Hit@5 of 90.11%, compared to 82.97% for TF-IDF cosine retrieval. However, the BM25-based classifier does not outperform the claim-only baseline on the evidence-bearing segment. The most plausible conclusion is that successful retrieval alone is no guarantee that scientific claims can be classified correctly with the lexical framework we evaluated.

The remaining comparison demonstrates a structural limitation that needs to be addressed in future evidence-based evaluations. The taxonomy-based analysis of 50 sampled errors provides an additional account of classification failures across the three conditions. The student reviewed and verified the classifications against the exact historical local corpus snapshot; however, the absence of an independent second annotator limits the assessment of inter-annotator agreement and should be addressed in future work.

12 Reproducibility and Artifact Traceability

The companion project repository contains source modules, scripts, YAML configs, prediction CSV, retrieval-rank files, metric JSON, confusion matrices, the error codebook, sampling metadata, a protocol-change log, and original run logs. Prediction files are time-stamped and their SHA-256 hashes are part of the evaluation metadata. The final package includes a read-only validator that verifies the hashes, the count of 286 rows, the unique claim identifiers and the equality of the A/B/C claim sets. The verifier was run successfully again by the final assessment. Model training was not re-run due to raw SciFact files not being found in the project archive.

The first environment snapshot lists Python 3.11.2 and a CPU-only version of Torch; no CUDA was available. There is no Colab execution documentation for the project. The autonomous review container ran on python 3.13.5 and did not have rank-bm25 installed. The collection could not be completed for the entire test suite because of two test modules that required rank-bm25==0.2.2 and prevented retrieval dependencies. 51 tests passed, except for those two modules; three RQ3 enrichment tests and five metadata-code tests also passed independently. This is partial test coverage on Python 3.13.5 not a full suite run on the documented Python 3.11.2 environment.

The archived project documentation shows that it passed 59 tests, but the full raw test stdout was not included in the source archive. This historical count was therefore derived from project reports and was not independently validated in this assessment. The upstream fetch configuration reveals a moving release or latest tarball. The file hashes are documented, but the upstream commit identifier is not. The licensing terms on the component level were retrieved from the official AllenAI repository: CC BY 4.0 for claims/evidence annotations, ODC-By 1.0 for S2ORC abstracts and Apache 2.0 for the repository code [12]. The downloaded archive is not tied to a specific upstream commit, which means that the archive and the revision license are not fully linked. Any redistribution of raw data requires prior license verification at the snapshot level, a fixed pin of the upstream revision, and a pristine clean-checkout reproduction.

Data and Code Availability

In the project repository we keep the code, scripts, configurations, archived output results and logs. No raw SciFact files and fitted model entities are included. The official AllenAI SciFact S3 release/latest archive lists the source of acquisition, retrieved on 30 October 2026; its SHA-256 hash can be found in the previous project log. The repository license under sanction is CC BY 4.0 for claims/evidence annotations, ODC-By 1.0 for S2ORC corpus abstracts, and Apache 2.0 for code [12]. Downloaded archive is referred to by its documented hash but is not linked to an immutable upstream commit. A different Hugging Face mirror has CC BY-NC 2.0 and a different split configuration, but this was not the source for these experiments. Raw data are not included until snapshot level verification is reached.

AI Assistance and Research Integrity

The project documentation describes how to use AI instruments for programming, problem solving and document creation. All numerical results reported in this paper have been verified against the time-stamped experimental records kept during the course of the project. In this revision, no additional model training or retrieval experiments were performed. The 50 sample entries used for RQ3 were analysed, and their error classifications were verified against the relevant project documentation, the corresponding condition-specific inputs, and the exact historical local corpus snapshot. The final interpretation of these classifications reflects the student's review and evaluation. Furthermore, the student took responsibility for the research decisions, analysis, interpretation, and reported results, and checked and confirmed the theoretical link between the initial literature review and the present experimental study.

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